

HDOC DAY – 21

QUESTION:

Write a menu-driven C program that accepts two positive integers m and n , where $m \leq n$. The program should repeatedly display a menu and use a switch statement to perform the selected operation:

1. Print the first 10 multiples of each integer from m to n .
2. Print the addition table of each integer from m to n , using addends from 1 to 10.
3. Print the subtraction table of each integer from m to n , using subtrahends from 1 to 10.
4. Divide each integer from m to n by integers 1 to 10 and display the results rounded to two decimal places.
5. Exit the program.

Algorithm:

1. Start.
2. Input two positive integers m and n .
3. Check whether $m \leq n$ and both numbers are positive.
4. Display the following menu:
 1. Multiplication Table
 2. Addition Table
 3. Subtraction Table
 4. Division Table
 5. Exit
5. Accept the user's choice.
6. Use a switch statement to perform the selected operation.
7. For choice 1, print the first 10 multiples of every integer from m to n .
8. For choice 2, print the addition table using numbers 1 to 10.
9. For choice 3, print the subtraction table using numbers 1 to 10.
10. For choice 4, divide each integer by numbers 1 to 10 and display the result up to 2 decimal places.
11. For choice 5, exit the program.
12. For any other choice, display "Invalid choice".
13. Repeat the menu until the user selects choice 5.
14. Stop.

Test Case	m	n	Menu Choice	Expected Result
1	2	3	1	Prints first 10 multiples of 2 and 3
2	2	3	2	Prints addition tables of 2 and 3
3	2	3	3	Prints subtraction tables of 2 and 3
4	2	3	4	Prints division tables of 2 and 3 up to 2 decimal places
5	2	3	5	Program exits

>C Code

```

1#include <stdio.h>
2
3int main()
4{
5    int m, n, choice, i, j;
6
7    printf("Enter two positive integers m and n (m ≤ n): ");
8    scanf("%d %d", &m, &n);
9
10   if (m ≤ 0 || n ≤ 0 || m > n)
11   {
12       printf("Invalid input.\n");
13       return 0;
14   }
15
16   do
17   {
18       printf("\nMENU\n");
19       printf("1. Multiplication Table\n");
20       printf("2. Addition Table\n");
21       printf("3. Subtraction Table\n");
22       printf("4. Division Table\n");
23       printf("5. Exit\n");
24
25       printf("Enter your choice: ");
26       scanf("%d", &choice);
27
28       switch (choice)
29       {
30           case 1:
31               for (i = m; i ≤ n; i++)
32               {
33                   printf("\nTable for %d:\n", i);
34                   for (j = 1; j ≤ 10; j++)
35                       printf("%d x %d = %d\n", i, j, i * j);
36                   break;
37               }
38           case 2:
39               for (i = m; i ≤ n; i++)
40               {
41                   printf("\nAddition table for %d:\n", i);
42                   for (j = 1; j ≤ 10; j++)
43                       printf("%d + %d = %d\n", i, j, i + j);
44                   break;
45               }
46           case 3:
47               for (i = m; i ≤ n; i++)
48               {
49                   printf("\nSubtraction table for %d:\n", i);
50                   for (j = 1; j ≤ 10; j++)
51                       printf("%d - %d = %d\n", i, j, i - j);
52                   break;
53               }
54           case 4:
55               for (i = m; i ≤ n; i++)
56               {
57                   printf("\nDivision table for %d:\n", i);
58                   for (j = 1; j ≤ 10; j++)
59                       printf("%d / %d = %.2f\n", i, j, (float)i / j);
60                   break;
61               }
62           case 5:
63               printf("Exiting program... \n");
64               break;
65           default:
66               printf("Invalid menu choice.\n");
67       }
68   } while (choice ≠ 5);
69
70   return 0;
71 }

```

>Execution and compilation:

*TEST CASE 1:

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ gcc day21.c -o day21
[priyadarshna@kali:~]$ ./day21
Enter two positive integers m and n (m <= n): 2 3

MENU
1. Multiplication Table
2. Addition Table
3. Subtraction Table
4. Division Table
5. Exit
Enter your choice: 1

Table for 2:
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Table for 3:
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

MENU
1. Multiplication Table
2. Addition Table
3. Subtraction Table
4. Division Table
5. Exit
Enter your choice: 
```

*TEST CASE 2:

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ gcc day21.c -o day21
[priyadarshna@kali:~]$ ./day21
Enter two positive integers m and n (m <= n): 2 3

MENU
1. Multiplication Table
2. Addition Table
3. Subtraction Table
4. Division Table
5. Exit
Enter your choice: 2

Addition table for 2:
2 + 1 = 3
2 + 2 = 4
2 + 3 = 5
2 + 4 = 6
2 + 5 = 7
2 + 6 = 8
2 + 7 = 9
2 + 8 = 10
2 + 9 = 11
2 + 10 = 12

Addition table for 3:
3 + 1 = 4
3 + 2 = 5
3 + 3 = 6
3 + 4 = 7
3 + 5 = 8
3 + 6 = 9
3 + 7 = 10
3 + 8 = 11
3 + 9 = 12
3 + 10 = 13

MENU
1. Multiplication Table
2. Addition Table
3. Subtraction Table
4. Division Table
5. Exit
Enter your choice: 
```

*TEST CASE 3:

```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ gcc day21.c -o day21
[priyadarshna@kali:~]$ ./day21
Enter two positive integers m and n (m <= n): 2 3

MENU
1. Multiplication Table
2. Addition Table
3. Subtraction Table
4. Division Table
5. Exit
Enter your choice: 3

Subtraction table for 2:
2 - 1 = 1
2 - 2 = 0
2 - 3 = -1
2 - 4 = -2
2 - 5 = -3
2 - 6 = -4
2 - 7 = -5
2 - 8 = -6
2 - 9 = -7
2 - 10 = -8

Subtraction table for 3:
3 - 1 = 2
3 - 2 = 1
3 - 3 = 0
3 - 4 = -1
3 - 5 = -2
3 - 6 = -3
3 - 7 = -4
3 - 8 = -5
3 - 9 = -6
3 - 10 = -7

MENU
1. Multiplication Table
2. Addition Table
3. Subtraction Table
4. Division Table
5. Exit
Enter your choice: █
```

*TEST CASE 4:

```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
[priyadarshna@kali:~]$ gcc day21.c -o day21
[priyadarshna@kali:~]$ ./day21
Enter two positive integers m and n (m <= n): 2 3

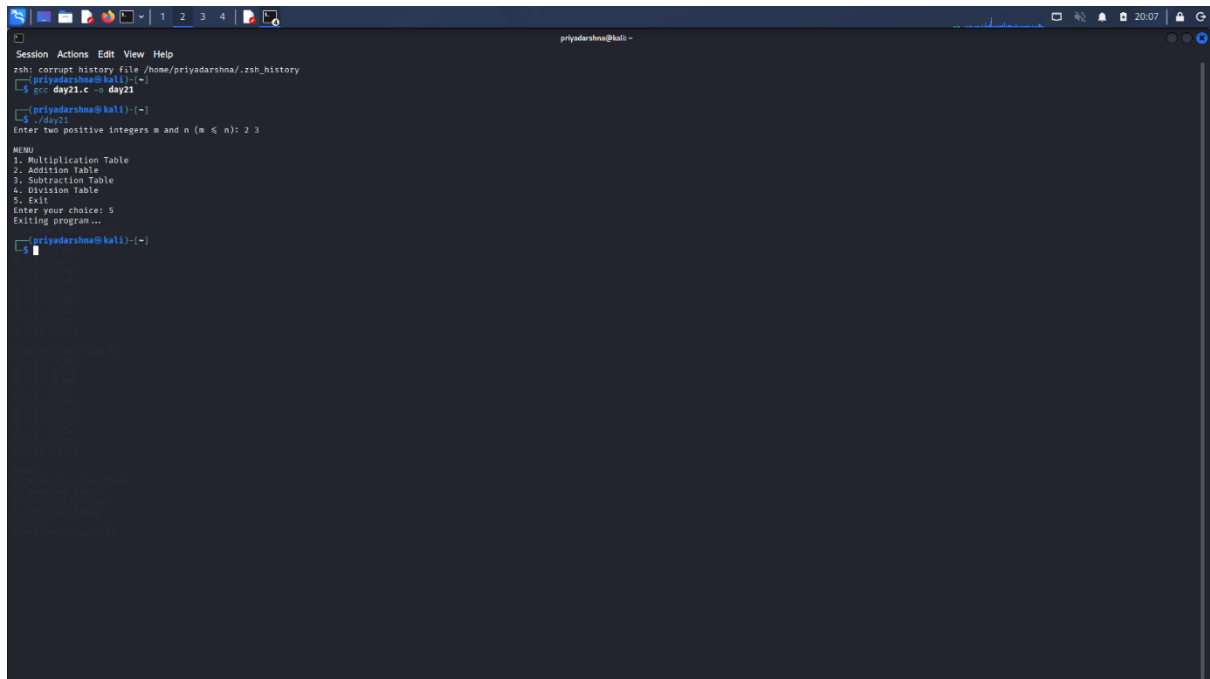
MENU
1. Multiplication Table
2. Addition Table
3. Subtraction Table
4. Division Table
5. Exit
Enter your choice: 4

Division table for 2:
2 / 1 = 2.00
2 / 2 = 1.00
2 / 3 = 0.67
2 / 4 = 0.50
2 / 5 = 0.40
2 / 6 = 0.33
2 / 7 = 0.29
2 / 8 = 0.25
2 / 9 = 0.22
2 / 10 = 0.20

Division table for 3:
3 / 1 = 3.00
3 / 2 = 1.50
3 / 3 = 1.00
3 / 4 = 0.75
3 / 5 = 0.60
3 / 6 = 0.50
3 / 7 = 0.43
3 / 8 = 0.38
3 / 9 = 0.33
3 / 10 = 0.30

MENU
1. Multiplication Table
2. Addition Table
3. Subtraction Table
4. Division Table
5. Exit
Enter your choice: █
```

***TEST CASE 5:**



```
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
priyadarshna@kali:~$ gcc day21.c -o day21
priyadarshna@kali:~$ ./day21
Enter two positive integers m and n (m <= n): 2 3

MENU
1. Multiplication Table
2. Addition Table
3. Subtraction Table
4. Division Table
5. Exit
Enter your choice: 5
Exiting program...
priyadarshna@kali:~$
```